Referee Report

Housing First or Treatment First?

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This paper aims to evaluate two proposed housing assistance programs, Housing First (HF) or Treatment First (TF). The author addresses the topic by developing a novel quantitative model incorporating housing and homelessness, drug experimentation and addiction, mortality, and labor markets within a general equilibrium, overlapping generations framework. The model parameters are calibrated to match key data facts in policy-relevant dimensions, aiming to align the magnitude, incidence, and interrelationships between homelessness, drug use, and other model features. The author then evaluates the model's ability to match other features of homelessness, drug-use, and income and wealth distribution in an untargeted fashion. Finally, the author uses the calibrated model to perform the key policy analysis around which the rest of the paper is written. The author implements HF and TF policies of varying generosities within the model, comparing allocations and welfare. The author finds that vouchers should be made universally available, rather than rationed, with an overall recommendation in favor of the universal-TF policy once consortium effects are taken into account.

This paper tackles an important and policy-relevant topic using an ambitious model, incorporating many important dimensions of the debate between these policies in an endogenous fashion. There are several details of the analysis in which the author has been very careful. At the same time, I have concerns about whether the current model, calibration, and policy analysis are sufficient to support the paper's main conclusions. I will discuss my concerns in detail below.

Main Comments

Note: I have ordered these comments in a way in which I feel is natural given the flow of the paper, not in order of importance. I would order these comments by importance in the following way: 2), 1), 4), 3).

1. **Model.** The topic surrounding the policy discussion is multifaceted, and the author has done an admirable job incorporating several of these dimensions into a single framework. That said, there are several modeling choices that either require further explanation or should be changed, and these have the potential to interact with the policy experiment in a non-trivial way.

   (a) Mental health is central to the economic and policy environment being studied, but it is largely absent from the model. Given the empirical connection between homelessness, addiction, labor-market outcomes, and mental health, the author should either incorporate mental health more explicitly or provide a much stronger argument for why excluding it does not materially affect the policy analysis. It seems to me that mental health contributing to drug use and addiction as well as homelessness would weaken TF dramatically as a policy.

   (b) One of the goals of TF as a policy is to encourage addicts to seek treatment for their addiction (it is even in the name of the policy!). However, this mechanism

is ruled out in the model by the assumption that recovery is random. I found the argument for ruling out any possibility of this mechanism to be fairly weak given its centrality to one of the two policies studied. That said, I expect the presence of this mechanism to strengthen TF as a policy relative to HF.

(c) The paper discusses drug-price responsiveness as evidence that drug use is a rational, forward-looking choice, and the calibration targets a participation elasticity with respect to drug prices. However, in the model, the drug price appears to be fixed at one and does not adjust in the policy counterfactuals. Given the framing of the paper as providing a general equilibrium framework through which to evaluate the effectiveness of the policy, and the strong potential for price responses to potentially offset reductions in drug use from TF policies, this seems like a strong assumption that either should be relaxed or better defended.

(d) The paper emphasizes mortality risk as the key mechanism generating drug use concentrated among low-income individuals. However, in a dynamic model with savings, borrowing constraints, and income risk, it seems possible that the labor-market and income-risk consequences of drug use could also generate a low-income concentration of drug use. The paper should demonstrate that the claimed mortality/value-of-life channel is actually the quantitatively important one. A simple way to do this would be to compare the relationship between income and drug use in the baseline model to an otherwise similar version of the model in which mortality is not affected by drug use. I would find this more intuitive than a toy example that abstracts from dynamic savings decisions.

(e) More generally, the appendix is useful for developing intuition, but I found it somewhat disorganized and narrower than the surrounding discussion suggests. The results are derived in very specific static settings, while the main model is dynamic and contains several interacting mechanisms. Some claims around the appendix are therefore more general than what the appendix actually establishes. In my opinion, the appendix should be reorganized and made more precise and the main text discussion should describe it more clearly as intuition rather than as a general proof of the model's mechanism.

2. **Calibration/Quantification.** As this paper is entirely quantitative in nature, the quantification is of utmost importance and for the audience to believe the conclusions regarding the policy analysis. However, I have some major concerns in this dimension.

(a) When the results of the calibration exercise is reported in Table 2, a notable omission is the model fit. That is, the author never reports the model's moments even though they would naturally fit in the table. The closest that the author comes to reporting these moments are by saying, "My benchmark closely matches all calibration targets." This is incredibly vague. The moments are close by what criteria? If they truly are close to the extent that the typical reader would agree so, then nothing is lost by the inclusion of the model moments within this table.

(b) This then relates to my next point. In Tables 3 and 4, the author reports untargeted moments in the data and in the model. I find these tables to

broadly demonstrating that the model misses substantially across the most important dimensions presented. This is problematic in and of itself - I will get to this point next. However, most problematic in my eyes is that the author frames the model as having done a reasonably good job of matching these moments. We are supposed to take away from this, that "The model's ability to replicate these non-targeted moments – particularly the concentration of drug use and homelessness among low-income and prime-age individuals – provides confidence that the counterfactual experiments capture the relevant behavioral margins through which HF and TF policies operate." Yet, this was not my takeaway. Given the context of the non-reported moments within the model from Table 2, this disagreement between my own and the author's interpretation of the quality of the non-targeted moment fits makes me question whether the fit of the targeted moments is as good as claimed. Without being able to verify myself through inspection of the model's moments in comparison with the data moments, I cannot help but question both the fit of the calibration and also the entire quantification exercise. This is the contribution of the paper, so losing confidence in the quantification is problematic.

(c) Back to the untargeted moments themselves, I think some of the misses are economically important rather than cosmetic:

   i. Table 3 (a) and (b): The model overstates the economic deprivation associated with homelessness and drug use. In the model, all homeless individuals are unemployed, compared with 40–52 percent in the data, and average income among the homeless is essentially zero, compared with 19.1 percent of average income in the data. Similarly, the average income of drug users is 41 percent of average income in the model, compared with 74 percent in the data. These are large differences in exactly the state variables that discipline the model's mechanism: income and employment determine the opportunity cost of drug use, the value of remaining alive, eligibility for assistance, tax payments, and the fiscal gains from moving people into employment. Because the policy comparison turns on the behavior and welfare of homeless individuals and drug users, a model in which these groups are much poorer and less attached to employment than in the data may evaluate HF and TF at the wrong margins.

   ii. Table 4 (b): The model predicts that homelessness should be associated much more strongly with younger individuals than in the data, and that agents should generally age out of homelessness. In the model, 73.1 percent of homeless individuals are between ages 26 and 44, compared with 48.1 percent in the California data. Conversely, only 9.3 percent of the model's homeless population is age 55 or older, compared with 27.1 percent in the data. This matters because the costs and benefits of housing assistance are not age-neutral in the model. Younger agents have more remaining working years and more time for addiction and housing transitions to compound, while older agents have different mortality risks, labor-market attachment, and fiscal consequences. If the model shifts homelessness toward younger ages, it may overstate the long-run labor-market and addiction-response channels while understating the role of policies for older homeless individuals.

(d) The model should get right not only the level of drug use and homelessness, but also the key behavioral elasticities that discipline the policy experiments. Relevant objects include the response of drug use to voucher generosity or conditionality, the response of homelessness to voucher access, the response of labor supply and employment to vouchers, and the response of recovery or addiction persistence to housing status. Some of these elasticities may not be directly observable. However, the RCT literature cited in the introduction may provide identified elasticities that the model can match in a targeted or untargeted fashion. This would be very convincing to me that the model matches reality along the most relevant dimensions.

(e) The closest current calibration target in this spirit is the elasticity of drug-use participation with respect to drug prices. While I would prefer the elasticities above to be matched instead, matching this elasticity seems useful. Unfortunately, I found the description of the calculation of this moment to be extremely vague, to the extent that I would not be able to replicate it. This should be clarified - I would appreciate an appendix describing the model definitions of the moments more thoroughly in general.

(f) I did find the identification explanation in Section 4.2 to be quite helpful. Given how important the quantification is to the structure of the paper, I would like this to also be shown numerically. This could be done in many ways - reporting the Jacobian of the moment vector with respect to the parameter vector, reporting the sensitivity statistic from Andrews, Gentzkow, and Shapiro (2017), through a figure showing how perturbing parameters locally around the baseline calibration affects the associated moments, etc. With the arguments in Section 4.2 and numerical evidence supporting this intuition, I would find the identification argument to be compelling.

- One caveat to this is that some of the data moments are not well measured - for example, take the elasticity of drug use with respect to price which has a range of plausible estimates from -0.36 to -0.9 depending on the drug. This moment also seems quite informative for the policy exercise. The chosen value to calibrate to for this moment surely matters for the policy analysis, and whether or not the results are robust to this choice (and any other choice of target for which the measurement is imprecise) should be demonstrated.

3. **Policy Analysis.** The policy analysis has several implementation details that should be addressed.

(a) The loss-of-consortium adjustment is important for the final welfare ranking, but it is not formally integrated into the model. If premature deaths impose welfare losses on other individuals, then these losses should in principle affect the value of life and agents' decisions (such as the decision to use drugs). Instead, the paper adds the loss-of-consortium effect only after solving the model. I find this to be an internal inconsistency. That said - this seems like something that can be incorporated directly into the model without adding much more computational burden given how random bequests are already handled.

(b) I find the adjustment used to be too ad-hoc. κ_{cons} was plucked out of thin air, yet makes an enormous impact on the final policy rankings. To play devil's advocate on the point that the chosen value was conservative, a truly conservative value for a parameter plucked out of thin air such as this would be zero, which results in a different policy recommendation.

(c) The fact that the loss-of-consortium adjustment changes the preferred policy is itself enough to suggest that the language around the policy results should be toned down a bit. Universal TF is not unambiguously preferred under all welfare criteria considered in the paper. In particular, the ranking between universal HF and universal TF depends on including this additional welfare adjustment. An example of language that should be softened is the conclusion's statement that the debate is resolved in favor of universal TF.

(d) The author should solve for proper transition paths. Since the model emphasizes that addiction adjusts slowly while housing outcomes respond more quickly, the dynamic transition is central to interpreting the policy experiments. This could, in principle, impact policy recommendations. I find the current strategy, of a several-period PE adjustment, to be a half-measure that is limited in its informativeness and opens up more questions than it answers.

4. **Sensitivity of Results** Given the narrow margins of the welfare comparison across models as well as the model's complexity, I find it unlikely that the results of the policy analysis are completely robust to some of the choices made in the model and calibration, including some choices which seem entirely arbitrary. Given the strength of the claims made about the results of the policy exercise, much more robustness needs to be demonstrated.

   (a) TF assumes that drug use is perfectly observed and perfectly enforceable. This seems difficult to implement in practice, even with drug testing. Drug users may evade detection, tests may be imperfect, and enforcement may be inconsistent.[1] If drug use were imperfectly observed, this would likely reduce the effectiveness of TF. The paper should analyze the sensitivity of the results to imperfect detection.

   (b) Relatedly, the paper should consider the consequences of false positives. If some non-users lose voucher access because of erroneous tests, then TF may impose substantial costs on individuals who are not violating the sobriety requirement. This could materially affect both welfare and homelessness outcomes.

   (c) The policy analysis should also account, at least in sensitivity exercises, for compliance costs, stigma, administrative burdens, and other implementation costs associated with voucher access. I found the argument that drug tests would cost only billions of dollars per year to implement to both be most likely a massive underestimation that ignored administrative overhead and assumed far more efficiency than is warranted of government policy, and also to be too large a number to entirely ignore in the first place.

[1] See, e.g., Coombs and Ryan (1990).

(d) The paper should report sensitivity to parameters that are critical for the policy analysis yet are difficult to identify such as the parameters governing addiction transitions, recovery probabilities, addiction-related mortality, homelessness-related mortality, and the employment consequences of addiction and homelessness.

Smaller Comments and typos

1. I do not understand why there is assumed to be an age at which drug experimentation will not occur. This seems inconsistent with Table 4(a), which shows U-shaped drug use by age.

2. The results tables would be easier to interpret if they included both percentage changes and baseline/counterfactual levels for the main policy outcomes.

3. Page 5, paragraph 2, word "well-aware": remove the hyphen. Correct to "well aware."

4. Equation (9) ends with a period that should be a comma, given the text that comes after.

5. Page 25, calibration target 19, sentence "Using data from New York City and Philadelphia, Kuhn and Culhane (1998), report...": remove the comma after the citation.

6. Page 28, footnote 7, phrase "The magnitude in our calibration": the paper is single-authored, change to singular.

7. Abramson citation has forthcoming listed twice.

8. Karlman et al. citation, author names "Kinneruda" and "Kragh-Sørensena" - neither name ends in an 'a'.

9. Goering et al. has "Guido. Powell ," which should probably be "Guido Powell," with no period after the first name.

10. Degenhardt et al. lists "Hamamd Ali." Hammad is misspelled.

References

[1] Andrews, Isaiah, Matthew Gentzkow, and Jesse M. Shapiro. 2017. "Measuring the Sensitivity of Parameter Estimates to Estimation Moments." *Quarterly Journal of Economics* 132(4): 1553–1592.

[2] Coombs, R. H., and F. J. Ryan. 1990. "Drug Testing Effectiveness in Identifying and Preventing Drug Use." *American Journal of Drug and Alcohol Abuse* 16(3–4): 173–184.